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Roll No

EC-403 (GS)
B.E. IV Semester Examination, June 2020
Grading System (GS)
Digital Electronics
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Convert the following in the radix shown:
 - i) $(153.6875)_{10} = ()_8$
 - ii) $(7621)_8 = ()_2$
 - iii) $(BA02:4A)_{16} = ()_4$
 - iv) $(D0A1.3A)_{16} = ()_{10}$
2. Subtract the following using 1's and 2's compliment
 - i) $(1010)_2$ from $(1111)_2$
 - ii) $(1010)_2$ from $(1000)_2$
3. State and explain the De-Morgan's theorems which convert a sum into a product form and Vice Versa.
4. Minimize the following function and realize using NAND gate.
$$F = ABC\bar{D} + \bar{A}BCD + \bar{A}\bar{B}CD + ABCD + A\bar{B}\bar{C}D + \bar{A}\bar{B}CD$$
5. a) Explain organization and construction of the following:
 - i) PAL
 - ii) ROMb) Write short notes on :
 - i) Interfacing
 - ii) CMOS
6. Explain the following:
 - a) Race around problem.
 - b) Synchronous and asynchronous counters.
 - c) Johnson and ring counter.

OR

Design D-flip-flop using T-flip -flop.
7. With a neat sketch, explain the operation of a bistable multivibrator.

OR

Explain the operation of the RTL NOR and NAND gate.
8. Answer any two of the following.
 - a) Prove the identity using Boolean algebra and De-Morgan's theorem:
$$\overline{AB + BC + CA} = \bar{A}\bar{B} + \bar{B}\bar{C} + \bar{C}\bar{A}$$
 - b) Draw and explain the 4-bit parallel binary adder.
 - c) Explain the synchronous 4-bit up-down counter.
 - d) Define fan-in, fan-out terms.